

Technology Stack

Date	15 July 2026
Team ID	SWTID-2026-8096
Project Name	Rising Waters
Maximum Marks	2 Marks

Technology Stack Details:

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript	Used to create a responsive and interactive user interface for entering rainfall data and displaying flood prediction results. CSS provides attractive styling and JavaScript improves user interaction.

2	Backend / Server-Side	Python, Flask	Flask handles HTTP requests, connects the frontend with the machine learning model, processes user inputs, and generates prediction results quickly with a lightweight framework.
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3	Database / Data Storage	CSV Dataset (Pandas)	The flood prediction model is trained using rainfall data stored in CSV format. Pandas efficiently loads, processes, and manages the dataset during model training.
4	Cloud / Hosting / Deployment	Render	Render provides free and reliable cloud hosting for Flask applications with simple GitHub integration, automatic deployment, and online accessibility.
5	Version Control & CI/CD	Git, GitHub	Git tracks code changes while GitHub stores the project repository, enables collaboration, version management, and automatic deployment through Render.

6	Third-Party APIs / Other Tools	Scikit-learn, NumPy, Joblib, Gunicorn	Scikit-learn is used to train the Random Forest model, NumPy performs numerical computations, Joblib saves and loads the trained model, and Gunicorn serves the Flask application in production.
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